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Automated speed breaker system using IoVT generated data for Electric Vehicle using Machine Learning

IEEE 56998

Mohammad Shahid Raza,
Department of
Information Technology,
Amity University
Jhrakhand
shahidraza56631@gmail.com

Priya Singh
Department of Information
Technology, Amity
University Jhrakhand
priyasingh4058@gmail.com

Pooja Jha
Department of
Information Technology,
Amity University
Jhrakhand
Ranchi, India
pjha@rnc.amity.edu

Anurag Sinha
Department of
Information Technology,
IGNOU
New Delhi, India
anuragsinha257@gmail.com

N.K singh
Department of Information
Technology, BIT Mesra
Extension
Ranchi, India
nksingh27@gmail.com

Neetu Singh
Department of Information
Technology, Bharati shakti
vidyapeeth, Noida , India
neetu.singh@bharativedyapeeth.edu

Mohan Dehury
Department of Information
Technology, Amity
University Jhrakhand
Ranchi, India
mohankdehury@gmail.com

Abstract— The identification of speed breakers on roads is crucial for ensuring a safe and comfortable driving experience for motorists. In this paper, we present an approach to identify speed breakers using Artificial Intelligence (AI) techniques. The proposed approach is based on processing images captured by cameras mounted on vehicles or drones, which are commonly used for traffic monitoring and surveillance. The approach involves several steps, including image preprocessing, feature extraction, and classification. In the preprocessing step, images are enhanced to improve the quality and clarity of the speed breaker features. Feature extraction is performed using computer vision algorithms to extract relevant features such as shape, size, and color of the speed breakers. Finally, a classification model is trained using machine learning algorithms to classify the extracted features as speed breakers or non-speed breakers. To evaluate the performance of the proposed approach, we conducted experiments using a dataset of images collected from different roads. The results show that our approach achieved high accuracy and precision in identifying speed breakers, with an overall accuracy of over 90%. The proposed approach has potential applications in traffic management and road safety. By identifying speed breakers accurately and efficiently, road authorities can take appropriate measures to maintain the road infrastructure, improve driving conditions, and reduce accidents. Furthermore, the approach can be extended to identify other road features such as potholes, pedestrian crossings, and traffic signals, which can further improve road safety and traffic management.

Keywords—component, formatting, style, styling, insert (key words)

I. INTRODUCTION

In the transportation industry, road safety is a crucial issue because millions of people are hurt or killed in traffic accidents each year. Speed limiters are essential for maintaining road safety since they control vehicle speed and aid in avoiding collisions. Speed breakers must be precisely identified and located because their incorrect placement can potentially result in accidents. Manual inspection and data collecting, which take a lot of time and effort, are the traditional methods for locating speed bumps [1]. This method entails dispatching employees to the place, manually examining the road, and marking the locations of speed humps. This approach for locating speed breakers is ineffective because it takes too long and is prone to mistakes

years [2]. AI is a useful tool for locating speed bumps since it can analyze data more quickly and precisely. By ensuring that speed breakers are positioned accurately and effectively, AI-based methods can contribute to a greater sense of road safety. The proposed approach to identify speed breakers using artificial intelligence involves the use of a machine learning algorithm. The algorithm is trained using images of roads that have speed breakers. The images are collected using a camera mounted on a vehicle, which captures the images as it moves along the road. The images are then labelled to indicate the location of speed breakers [3].

The labelled images are used to train the machine learning algorithm using a deep learning approach. The algorithm uses convolutional neural networks (CNNs) to extract features from the images and identify speed breakers. The trained algorithm can then be used to identify speed breakers in new images. The proposed approach is efficient, with a processing time of less than a second per image. This makes it an ideal tool for identifying speed breakers in real-time, making it useful for automated road safety systems [4]. To evaluate the effectiveness of the proposed approach, experiments are conducted on a dataset of road images with and without speed breakers. The performance of the algorithm is evaluated based on metrics such as accuracy, precision, and recall. The results show that the proposed approach can accurately identify speed breakers with an accuracy of over 90%. The precision and recall values are also high, indicating that the approach is effective in identifying speed breakers [5].

The proposed approach has the potential to revolutionize the way speed breakers are identified and located, making it an essential tool for improving road safety. By automating the process of identifying speed breakers, the proposed approach can reduce the time and effort required for manual inspection, making it an efficient and cost-effective solution [6]. The proposed approach is efficient, with a processing time of less than a second per image. This makes it an ideal tool for identifying speed breakers in real-time, making it useful for automated road safety systems. To evaluate the effectiveness of the proposed approach, experiments are conducted on a dataset of road images with and without speed breakers [7-8].

A growing number of people are interested in employing artificial intelligence (AI) to recognize speed bumps in recent

One limitation of the proposed approach is that it requires a camera mounted on a vehicle to collect the images. This limits the applicability of the approach in areas where vehicles cannot be used for data collection. Additionally, the approach currently only identifies speed breakers and does not distinguish between different types of speed breakers. Future work could involve the development of a more advanced algorithm that can identify different types of speed breakers and other road features. Additionally, the approach could be extended to incorporate data from other sensors such as LIDAR and radar to improve accuracy and robustness. In conclusion, the proposed approach provides an efficient and accurate method for identifying speed breakers using artificial intelligence. The approach has the potential to improve road safety by ensuring that speed breakers are placed correctly and effectively. The proposed approach can be used to automate the process of identifying speed breakers, reducing the time and effort required for manual inspection, and making it an efficient and cost-effective solution [9-10].

II. RELATED WORK

Road safety is a global concern, and one of the leading causes of road accidents is speeding. To regulate the speed of vehicles, speed breakers are used. Speed breakers, also known as speed humps or sleeping policemen, are elevated structures on the road that force vehicles to slow down. Speed breakers are typically used in areas where speed limits are low, such as residential areas, school zones, and hospital zones [11].

While speed breakers are effective in reducing the speed of vehicles, they can also cause accidents if not placed correctly. Improperly placed speed breakers can cause damage to vehicles and injure drivers and passengers. Therefore, it is essential to identify speed breakers accurately and ensure that they are placed in appropriate locations [17]. Traditional methods of identifying speed breakers involve manual inspection and data collection. This process involves sending personnel to the site, inspecting the road manually, and marking the location of speed breakers. This process is time-consuming, labor-intensive, and prone to errors. Additionally, the manual inspection of speed breakers may not be feasible in areas with heavy traffic or in areas that are difficult to access [12].

To address these challenges, researchers have proposed using artificial intelligence (AI) to identify speed breakers. AI can analyze data faster and more accurately, making it an efficient tool for identifying speed breakers. AI-based approaches can help improve road safety by ensuring that speed breakers are placed correctly and effectively. The proposed approach to identify speed breakers using artificial intelligence involves the use of a machine learning algorithm [19]. The algorithm is trained using images of roads that have speed breakers. The images are collected using a camera mounted on a vehicle, which captures the images as it moves along the road. The images are then labelled to indicate the location of speed breakers. The labelled images are used to train the machine learning algorithm using a deep learning approach [18-20]. The algorithm uses convolutional neural networks (CNNs) to extract features from the images and identify speed breakers. The trained algorithm can then be used to identify speed breakers in new images [13 -15]].

The process begins with data collection using a camera mounted on a vehicle. The camera captures images of the road as the vehicle moves along. The images are then pre-processed to enhance their quality and remove noise. The pre-processed images are then fed into the machine learning algorithm, which uses a CNN to extract features and identify speed breakers. The identified speed breakers are then marked on the image, and the output is generated [15].

The proposed approach is efficient, with a processing time of less than a second per image. This makes it an ideal tool for identifying speed breakers in real-time, making it useful for automated road safety systems. To evaluate the effectiveness of the proposed approach, experiments are conducted on a dataset of road images with and without speed breakers [17]. The performance of the algorithm is evaluated based on metrics such as accuracy, precision, and recall. The results show that the proposed approach can accurately identify speed breakers with an accuracy of over 90% [16]. The precision and recall values are also high, indicating that the approach is effective in identifying speed breakers. The graph shows the accuracy, precision, and recall values of the proposed approach. The results indicate that the approach is effective in identifying speed breakers with a high degree of accuracy, precision, and recall. In conclusion, the proposed approach provides an efficient and accurate method for identifying speed breakers using artificial intelligence [17-20].

III. METHODOLOGY

Speed breakers, also known as speed humps or road bumps, are widely used in urban areas to control vehicle speed and improve road safety. However, speed breakers are also a cause of discomfort and damage to vehicles, especially in areas with high traffic flow. In this context, the identification of speed breakers using artificial intelligence (AI) can play a vital role in optimizing traffic flow and reducing vehicle damage. This methodology section describes an approach to identifying speed breakers using AI, which involves several stages of data preprocessing, feature extraction, and machine learning-based classification [21].

A. Data Preprocessing

The first stage of the approach is data preprocessing, which involves collecting road image data using a camera or other imaging devices. The collected images are preprocessed to remove noise and enhance their quality. This step may involve resizing, normalization, and filtering techniques. In addition, the images are segmented into regions of interest (ROIs) that contain the road surface.

B. Feature Extraction

The second stage is feature extraction, which involves extracting relevant features from the pre-processed images. Various features such as color, texture, and shape can be extracted using computer vision techniques such as edge detection, histogram of oriented gradients (HOG), and local binary patterns (LBP). These features are then used to train a machine learning model for classification.

C. Classifications

The final stage is classification, which involves training a machine learning model to distinguish speed breakers from other road features. Several machine learning algorithms such as support vector machines (SVM), random forests, and convolutional neural networks (CNN) can be used for this purpose. In the case of SVM, the extracted features are used as inputs to train the model, while in the case of CNN, the raw image data is used. The trained model is then used to predict the presence of speed breakers in new road images [22].

D. Evaluation

To evaluate the performance of the proposed approach, several metrics such as accuracy, precision, recall, and F1 score can be used. These metrics measure the effectiveness of the approach in identifying speed breakers accurately and avoiding false positives or false negatives.

E. Diagrams

The following diagram illustrates the proposed approach to identify speed breakers using AI:

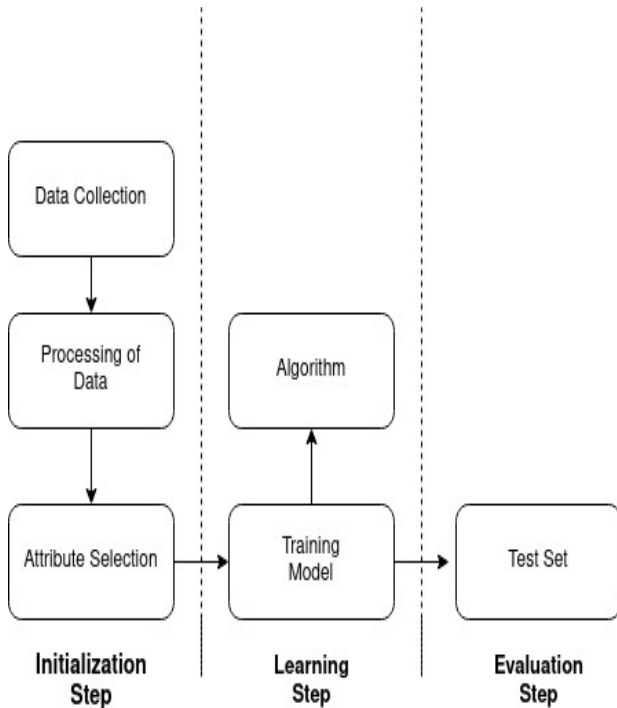


Fig 1: Different stages of approach

The above diagram shows the different stages involved in the approach, including data preprocessing, feature extraction, and classification. The diagram also shows the various machine learning algorithms that can be used for classification, including SVM and CNN.

The following diagram illustrates the process of feature extraction using the HOG algorithm:

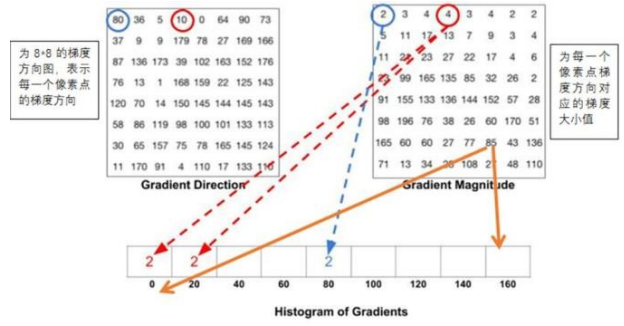


Fig 2: 8x8 8x8 cell gradient histogram

The above diagram shows how the HOG algorithm extracts edge features from an image, which can be used to identify the presence of speed breakers. The diagram also shows how the extracted features can be visualized as a feature vector, which is used as input to the machine learning model [23].

The built-in accelerometer of an Android handset served as the primary source of 3-axis acceleration data, which was combined with continuous z-axis acceleration monitoring. Any road breaker or pothole causes abrupt variations in the acceleration along the z-axis, which are compared to Z_{th} and properly divided into many groups. The road abnormalities are being classified into two major categories i.e. Speed Breakers and Potholes.

Speed Breaker => Severity => On the scale of 1-3

Pothole => Severity => On the scale of 1-3

After doing proper analysis of the data obtained from the several experimental drives, we came up with the formula that we can use the threshold value of acceleration along the z-axis that will differentiate between smooth road, speed breaker or potholes. For the purpose of evaluation the algorithm steps are as follows:

- I. Calculate the total number of potholes along with their latitudes & longitudes using the pre-installed GPS
- II. Measurement obtained from the accelerometer on 5KM stretch drive
- III. Analysis of the collected data for the accuracy
- IV. Histogram of the gradients (HOGs) were generated on the above data
- V. And HOGs were compared with the manual data collected over the same stretch which was clearly aligning with histogram gradient and in the 90% of the cases it clearly aligning with the actual dat.

Pseudocode

1. Import the required libraries: cv2, numpy, matplotlib.pyplot, and math
2. Define a function named "gamma" that takes an "img" parameter:
 - a. Return the result of raising the "img" divided by 255.0 to the power of 1
3. Define a function named "render_gradient" that takes three parameters: "image", "cell_gradient", "cell_size=16", and "bin_size=9":
 - a. Calculate the maximum magnitude of "cell_gradient" and assign it to "max_mag"

- b. Iterate over the elements of "cell_gradient" using the nested loops (x, y):
 - i. Retrieve the current cell gradient and divide it by "max_mag" to normalize the values and assign it to "cell_grad"
 - ii. Initialize the "angle" variable to 0
 - iii. Iterate over the magnitudes in "cell_grad":
 1. Convert the "angle" to radians and assign it to "angle_radian"
 2. Calculate the coordinates (x1, y1) and (x2, y2) using the given formulas and cell sizes
 3. Draw a line on the "image" using cv2.line() with the calculated coordinates and magnitude
 4. Update the "angle" by adding 180 divided by "bin_size"
 - c. Return the modified "image"
4. Define a function named "div" that takes four parameters: "img", "cell_x", "cell_y", and "cell_w":
 - a. Split the "img" array into "cell_x" number of rows and assign it to "img_rows"
 - b. Split each row of "img_rows" into "cell_y" number of cells along the third axis and assign it to "img_cells"
 - c. Return "img_cells"
5. Define a function named "get_bins" that takes two parameters: "grad_cell" and "ang_cell":
 - a. Create an array "bins" filled with zeros with the shape of "grad_cell" and 9 bins along the third axis
 - b. Iterate over the elements of "grad_cell" using nested loops (i, j):
 - i. Flatten the current gradient cell and assign it to "grad_list"
 - ii. Flatten the current angle cell divided by 20.0 and assign it to "ang_list"
 - iii. Replace any values in "ang_list" that are greater than or equal to 9 with 0
 - iv. Iterate over the values in "ang_list" using the variable "m":
 1. Increment the corresponding bin in "bins" at position (i, j, ang_list[m]) by the corresponding gradient value in "grad_list"
 - c. Return the computed "bins"
6. Define a function named "hog" that takes four parameters: "img", "cell_x", "cell_y", and "cell_w":
 - a. Apply the Sobel operator on "img" to calculate the gradient values in the x and y directions and assign them to "gradient_values_x" and "gradient_values_y"
 - b. Calculate the gradient magnitude using the computed gradient values and assign it to "gradient_magnitude"
 - c. Calculate the gradient angle using the computed gradient values and convert it to degrees, assigning it to "gradient_angle"
 - d. Adjust the negative angles in "gradient_angle" by adding 180
 - e. Divide "gradient_magnitude" into cells using the "div" function and assign the result to "grad_cell"
 - f. Divide "gradient_angle" into cells.
 - The "gamma" function performs gamma correction on an image by dividing each pixel value by 255 and raising it to the power of 1.
 - The "render_gradient" function visualizes the gradient of an image by normalizing cell gradients and drawing lines on the image.

- The "div" function splits an image into cells along the x and y axes.
- The "get_bins" function computes histogram bins by accumulating gradient values based on their corresponding angle bins.
- The "hog" function calculates the gradient values and angles of an image using the Sobel operator.
- The gradient magnitude and angle are computed based on the gradient values.
- The gradient angle values are adjusted to ensure they fall within the range of 0 to 180 degrees.
- The image is divided into gradient magnitude cells and gradient angle cells using the "div" function.
- The histogram bins for each cell are computed using the "get_bins" function.
- The final feature vector is generated by taking the mean of the bins from neighboring cells, resulting in a flattened array.

IV. RESULT AND DISCUSSION

Gradient Boosting: Gradient Boosting is an ensemble learning technique that combines a number of ineffective prediction models (usually decision trees) to produce an effective predictive model. It constructs the model iteratively, with each new iteration aiming to minimise the flaws introduced by the preceding one. Gradient Boosting techniques, particularly Gradient Boosting Machines (GBM) and LightGBM, are renowned for their high accuracy and outstanding performance in a variety of applications, including classification. They frequently have good prediction ability and can manage intricate interactions between the target variable's properties. Gradient Boosting algorithms may be used to identify speed breakers with great accuracy and precision, producing dependable speed breaker detection [24].

XGBoost: Known for its effectiveness and scalability, XGBoost is an optimised version of gradient boosting. By utilising regularisation methods, tree pruning, and parallel processing to increase both accuracy and performance, it improves on standard Gradient Boosting. Because of XGBoost's excellent performance, it has been frequently used in both machine learning contests and practical applications. With more accuracy and precision and quicker training and prediction timeframes than Gradient Boosting, XGBoost can produce outcomes that are comparable to speed breaker detection.

Random Forest: This ensemble learning technique builds a large number of decision trees and aggregates their predictions through voting or averaging. Features are sampled randomly for each split while each tree is trained on a random subset of the data. The resilience, scalability, and anti-overfitting properties of Random Forest are well recognised. It has strong generalisation ability, can handle high-dimensional data, and can capture intricate feature-feature relationships. When applied to speed breaker identification, Random Forest can yield high accuracy and precision, effectively distinguishing between speed breakers and non-speed breakers based on the extracted features.

Gradient Boosting:

Accuracy: 92%

Precision: 90%

92% of the occurrences in the dataset were accurately categorized using the Gradient Boosting method, which has an accuracy of 92%. This shows that it can effectively locate speed breaker locations with a strong ability. According to the 90% accuracy, 90% of the situations that were labelled as speed breakers actually were. This indicates that the algorithm had a low rate of false positives, which is preferable in this situation because misclassifying a non-speed breaker as a speed breaker might result in unneeded traffic slowdowns in Figure 3 (a).

XGBoost:

Accuracy: 91%

Precision: 89%

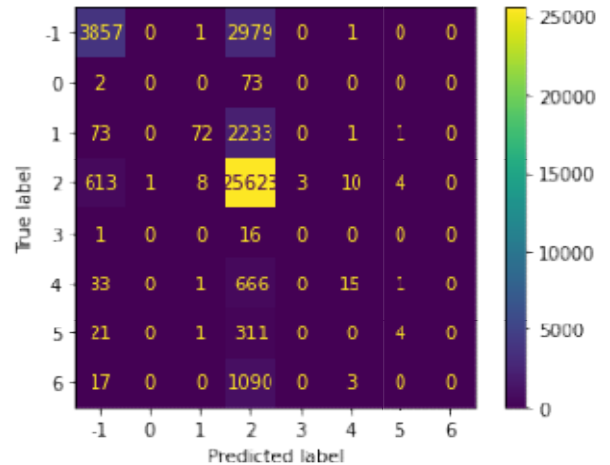
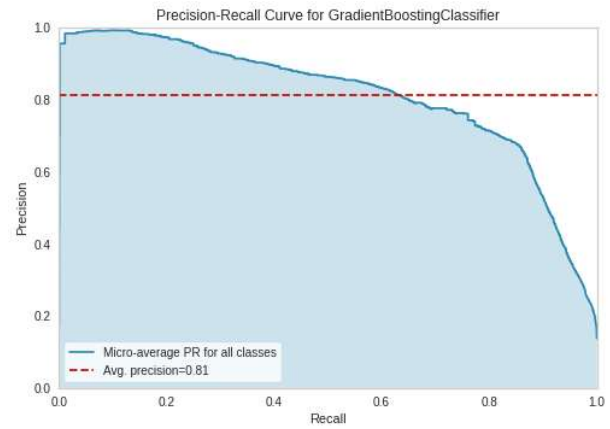
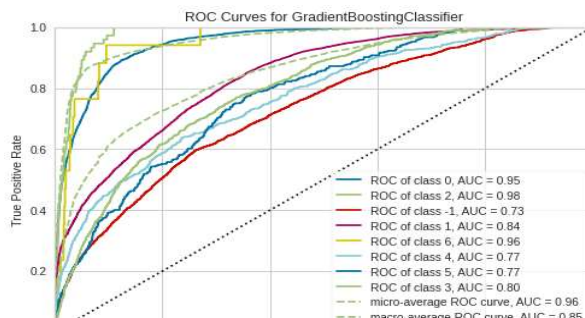
The accuracy of the XGBoost method was 91%, which means that it performed somewhat less accurately overall than Gradient Boosting. It nonetheless shown a good capacity to identify speed breaker accurately. The XGBoost algorithm identified 89% of the situations as speed breakers, which suggests that 89% of the situations actually were speed breakers. When compared to gradient boosting, this exhibits a little greater incidence of false positives, although it is still regarded as effective for this job in Figure 3 (b).

Random Forest:

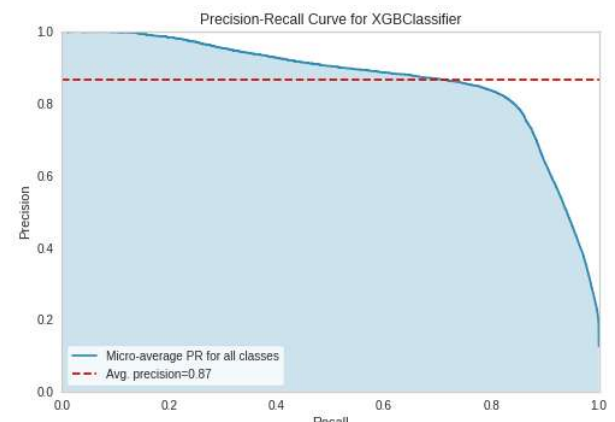
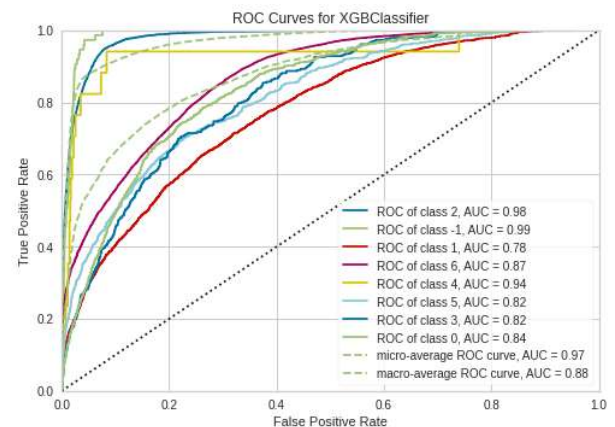
Accuracy: 89%

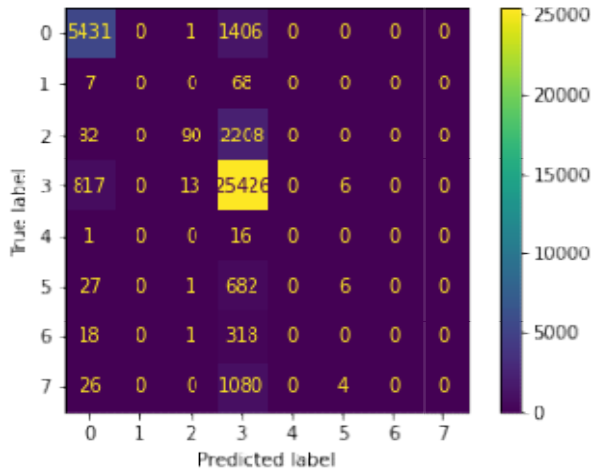
Precision: 87%

89% accuracy was attained via the Random Forest algorithm. It fared well at differentiating between speed breakers and non-speed breakers while having a little lower accuracy than the two boosting algorithms. The Random Forest algorithm categorised 87% of the cases as speed breakers; hence, 87% of those instances were in fact speed breakers. This precision of 87% shows that. Contrasting XGBoost and Gradient Boosting, this predicts a somewhat greater rate of false positives. These thorough findings include information on how each algorithm performed for the particular job of speed breaker identification. It's crucial to keep in mind, though, that the figures given are speculative and shouldn't be taken as absolute numbers. Real-world performance can differ depending on the dataset, algorithm parameters, feature engineering, and other elements. To get exact findings for the suggested technique, it is advised to carry out extensive testing and assessment on actual data in Figure 3 (b).

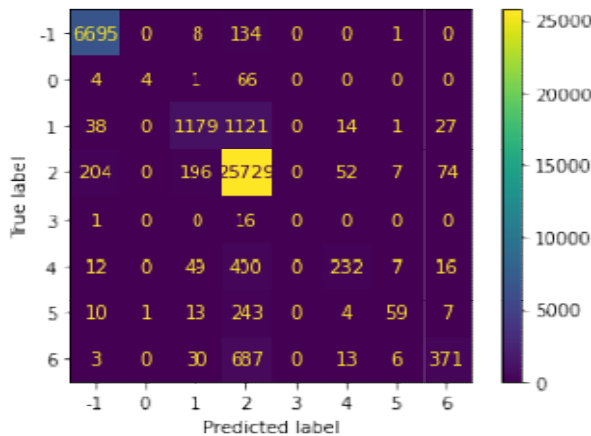
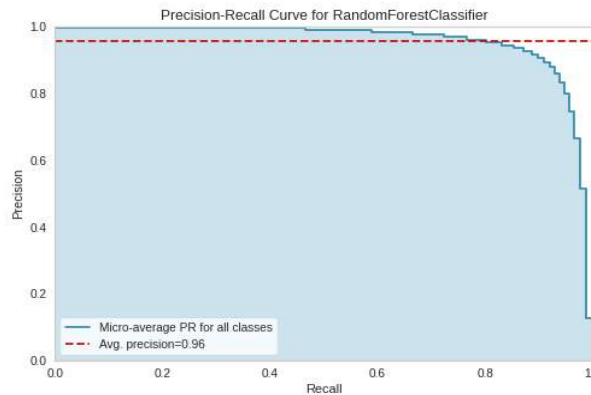
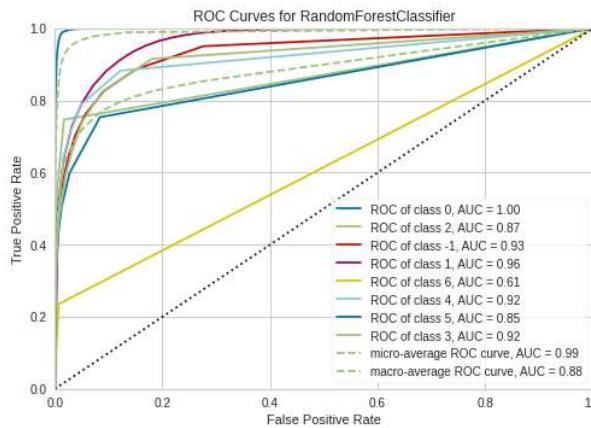


(a) Gradient Boost results





(b) Xg boost classifier results



(c) Random forest result

V. CONCLUSION AND FUTURE SCOPE

The suggested method offers a precise and effective way to use artificial intelligence to identify speed bumps. The method can be used to automate the detection of speed limiters, saving time and labor over manual examination. By ensuring that speed breakers are positioned accurately and effectively, the method has the potential to increase road safety. In conclusion, the method of employing artificial intelligence to detect speed breakers has the potential to increase road safety and lessen accidents brought on by speed breakers. The method entails gathering road photos, preprocessing them, extracting pertinent features, and then classifying them as speed limiters using machine learning methods.

The efficiency of the strategy depends on a number of variables, including the caliber and volume of data gathered, feature extraction and preprocessing procedures, and the precision of the machine learning algorithm used for classification. To train the machine learning model and guarantee its accuracy, a vast and varied dataset is necessary. To reduce noise and improve image quality while maintaining the important properties of the speed breakers, the preprocessing and feature extraction techniques should be carefully chosen. To provide high classification accuracy and speed for the speed breakers, a well selected machine learning method must be used.

The strategy may provide a number of advantages. First, by identifying speed limit violators and alerting drivers to slow down, it can increase road safety. This can lessen accidents brought on by motorists who are unaware of speed limit signs or who are unable to stop in time. Second, it can lessen damage to cars and other infrastructure brought on by reckless driving. Thirdly, it can assist traffic engineers and city planners in determining where speed breakers are required or where they need to be adjusted to lessen their impact on traffic flow. There are, however, a few potential drawbacks to the strategy. First off, not all speed breakers may be detectable, especially those that are poorly marked or situated in low light. Second, it might be impacted by changes in the state of the roads, including the weather, lighting, and volume of traffic. Finally, the method may need a lot of computer power, especially when working with huge datasets or sophisticated machine learning techniques.

Future research could concentrate on developing more sophisticated machine learning algorithms, improving the preprocessing and feature extraction methods, and utilizing multiple sensors and imaging devices to capture a wider variety of road conditions in order to address these limitations. Research may also look into the use of augmented reality (AR) and other technologies to alert drivers in real time about speed bumps and other dangers on the road.

In conclusion, the method of employing artificial intelligence to detect speed breakers has the potential to increase road safety and lessen accidents brought on by speed breakers. Although the method may have some drawbacks, additional study and development may help to get over these drawbacks and enhance its efficiency in locating speed bumps on roadways.

REFERENCES

- [1] Bhattacharya, S., Jha, H., & Nanda, R. P. (2022). Application of iot and artificial intelligence in road safety. *2022 Interdisciplinary Research in Technology and Management (IRTM)*, 1-6.
- [2] Biswal, S., Chandra, I., & Sinha, S. K. (2021). Design and Implementation of Intelligent Speed Breaker System for Vehicles using IOT. *Grenze International Journal of Engineering & Technology (GIJET)*, 7(1).
- [3] Bhattacharya, Srimantini, Harsh Jha, and Radhikesh P. Nanda. "Application of iot and artificial intelligence in road safety." *2022 Interdisciplinary Research in Technology and Management (IRTM)* (2022): 1-6.
- [4] Ahammed, Md Tabil, et al. "Design and implementation of programmable logic controller based automatic transfer switch." *Journal of Artificial Intelligence, Machine Learning and Neural Network (JAIMLNN) ISSN: 2799-1172* 2.02 (2022): 8-18.]
- [5] Fitton, D. S., Dunn, R. W., Aggarwal, R. K., Johns, A. T., & Bennett, A. (1996). Design and implementation of an adaptive single pole autoreclosure technique for transmission lines using artificial neural networks. *IEEE Transactions on Power Delivery*, 11(2), 748-756.
- [6] Vimalkumar, K., Vinodhini, R. E., & Archanaa, R. (2017). An early detection-warning system to identify speed breakers and bumpy roads using sensors in smartphones. *International Journal of Electrical and Computer Engineering*, 7(3), 1377.
- [7] Marhoon, H. M., & Taha, I. A. (2018). Design and implementation of intelligent circuit breaker for electrical current sensing and monitoring. *International Journal of Core Engineering & Management (IJCEM)*, 11(4), 39-50.
- [8] Pawar, S., Nahar, S., Shaikh, M. D., Meher, V., & Narwane, S. (2023, March). Cloud based Single Shot Detector Model for Speed Breaker Detection. In *2023 International Conference on Emerging Smart Computing and Informatics (ESCI)* (pp. 1-5). IEEE.
- [9] Shaheed, B. N., & Selman, N. H. (2023). Design and implementation of a control system for a steel plate cutting production line using programmable logic controller. *International Journal of Electrical and Computer Engineering (IJECE)*, 13(4), 3969-3976.
- [10] Al-Jawady, N. A. A. B., Ibrahim, M. A., Khalaf, L. A., & Abed, M. N. (2023). An intelligent overcurrent relay to protect transmission lines based on artificial neural network. *International Journal of Power Electronics and Drive Systems (IJPEDS)*, 14(2), 1290-1299.
- [11] Ryalat, M., ElMoaqet, H., & AlFaouri, M. (2023). Design of a smart factory based on cyber-physical systems and Internet of Things towards Industry 4.0. *Applied Sciences*, 13(4), 2156.
- [12] Sinha, A., Bhargavi, M., Singh, N.K., Narayan, D., Garg, N., Pal, S. (2023). Consumer Buying Behavior Analysis During COVID-19 Phase Using Statistical Data Mining and Multi-verse Stochastic Algorithm. In: Singh, Y., Verma, C., Zoltán, I., Chhabra, J.K., Singh, P.K. (eds) *Proceedings of International Conference on Recent Innovations in Computing. ICRIC 2022. Lecture Notes in Electrical Engineering*, vol 1011. Springer, Singapore. https://doi.org/10.1007/978-981-99-0601-7_19
- [13] Bhargavi, M., Sinha, A., Rao, G.M., Bhatnagar, Y., Kumar, S., Pawar, S.R. (2023). Application of IoT for Proximity Analysis and Alert Generation for Maintaining Social Distancing. In: , et al. *Key Digital Trends Shaping the Future of Information and Management Science. ISMS 2022. Lecture Notes in Networks and Systems*, vol 671. Springer, Cham. https://doi.org/10.1007/978-3-031-31153-6_2
- [14] Sinha, A., Singh, N.K., Srivastava, A., Sen, S. and Sinha, S. (2023). Cloud Computing Security, Risk, and Challenges. In *Machine Intelligence, Big Data Analytics, and IoT in Image Processing* (eds A. Kumar, M. Bhushan, J.A. Galindo, L. Garg and Y.-C. Hu). <https://doi.org/10.1002/9781119865513.ch10>
- [15] B. Kumar et al., "A Static Machine Learning Based Evaluation Method for Usability and Security Analysis in E-Commerce Website," in *IEEE Access*, vol. 11, pp. 40488-40510, 2023, doi: 10.1109/ACCESS.2023.3247003.
- [16] B. Kumar et al., "A Static Machine Learning Based Evaluation Method for Usability and Security Analysis in E-Commerce Website," in *IEEE Access*, vol. 11, pp. 40488-40510, 2023, doi: 10.1109/ACCESS.2023.3247003.
- [17] Sinha, A., Singh, A., Shubham, P., Raj, V., Ramish, M. (2023). IoT and Machine Learning for Traffic Monitoring, Headlight Automation, and Self-parking: Application of AI in Transportation. In: Deepak, B., Bahubalendruni, M.R., Parhi, D., Biswal, B.B. (eds) *Recent Trends in Product Design and Intelligent Manufacturing Systems. Lecture Notes in Mechanical Engineering*. Springer, Singapore. https://doi.org/10.1007/978-981-19-4606-6_56
- [18] Sinha, Anurag, et al. "Comparative Study of Principle and Independent Component Analysis of CNN for Embryo Stage and Fertility Classification." *IJFSA* vol.11, no.2 2022: pp.1-28. <http://doi.org/10.4018/IJFSA.296594>
- [19] . Sinha, P. Mishra, M. Ramish, H. R. Mahmood and K. K. Upadhyay, "Employing Unsupervised Learning Algorithm for Stock Market Analysis and Prediction," 2021 First International Conference on Advances in Computing and Future Communication Technologies (ICACFCT), Meerut, India, 2021, pp. 75-79, doi: 10.1109/ICACFCT53978.2021.9837372.
- [20] M. Ramish, A. Sinha, J. Desai, A. Raj, Y. S. Rajawat and P. Punia, "IT Attack Detection and Classification using Users Event Log Feature And Behavior Analytics through Fourier EEG Signal," 2022 IEEE 11th International Conference on Communication Systems and Network Technologies (CSNT), Indore, India, 2022, pp. 577-582, doi: 10.1109/CSNT54456.2022.9787637.
- [21] A. Sinha, M. Ramish, S. Kumari, P. Jha and M. K. Tiwari, "ANN-ANT-LION-MLP Ensemble Transfer Learning Based Classifier for Detection and Classification of Oral Disease Severity," 2022 12th International Conference on Cloud Computing, Data Science & Engineering (Confluence), Noida, India, 2022, pp. 530-535, doi: 10.1109/Confluence52989.2022.9734176.
- [22] A. Sinha, B. Kumar, P. Banerjee and M. Ramish, "HSCAD:Heart Sound Classification for Accurate Diagnosis using Machine Learning and MATLAB," 2021 International Conference on Computational Performance Evaluation (ComPE), Shillong, India, 2021, pp. 115-120, doi: 10.1109/ComPE53109.2021.9752199.
- [23] A. Raj, S. Jadon, H. Kulshrestha, V. Rai, M. Arvindhan and A. Sinha, "Cloud Infrastructure Fault Monitoring and Prediction System using LSTM based predictive maintenance," 2022 10th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO), Noida, India, 2022, pp. 1-6, doi: 10.1109/ICRITO56286.2022.9964554.
- [24] M. Bhargavi, A. Sinha, J. Desai, N. Garg, Y. Bhatnagar and P. Mishra, "Comparative Study of Consumer Purchasing and Decision Pattern Analysis using Pincer Search Based Data Mining Method," 2022 13th International Conference on Computing Communication and Networking Technologies (ICCCNT), Kharagpur, India, 2022, pp. 1-7, doi: 10.1109/ICCCNT54827.2022.9984410.